

Eric Gu

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Education

University of California, San Diego

BS/MS in Computer Engineering

June 2027

GPA: 3.8 | Major GPA: 4.0

- **Relevant Coursework:** DSA, Operating Systems, Software Engineering, Deep Learning Models, Reinforcement Learning

Experience

Incoming Software Development Engineer Intern

September 2026

Amazon – Irvine, CA – Device OS

Software Engineering Intern

June 2026 – September 2026

Viasat – Carlsbad, CA

- Built an LLM operations assistant using Google's ADK and Gemini (Vertex AI): a coordinator agent delegating to four MCP sub-agents and a Vertex AI Search base over 14,000+ Confluence pages.
- Integrated four MCP servers via ADK's McpToolset with least-privilege access (43-tool read-only GitHub allowlist) and pulled all runtime secrets from HashiCorp Vault via keyless GCP-IAM auth.
- Authored custom Grafana tools that interpolate BigQuery-backed panel SQL for live metrics, and deployed the agent as a private Cloud Run service with a FastAPI and single-page chat UI.

Software Development Intern

June 2023 – September 2025

MaxNorm LLC – Jersey City, NJ

- Researched cryptocurrency markets and blockchain protocols, analyzing trends with Python technical and fundamental methods to produce reports that informed web3 investment decisions.
- Developed and deployed Solidity smart contracts for DeFi prototypes on Ethereum testnets, reducing security vulnerabilities by 40% through rigorous simulation testing.
- Backtested quantitative trading strategies with Pandas and NumPy, evaluating Sharpe ratio and drawdown on historical data to refine algorithms reaching up to 25% simulated returns.

Projects

CUDA-NN | C++, CUDA

github.com/fordatab/gemm

- Beat PyTorch's own convolution path end-to-end (172.9 vs 181.9 ms whole-net) with a C++/CUDA extension swapping in a custom first-layer kernel — conv2-4 unchanged, so the delta is provably the kernel.
- Cut the single-channel conv 20% below autotuned FP32 cuDNN (35.3 vs 44.2 ms) with one kernel replacing cuDNN's three, folding bias into the accumulator to reach the memory floor.
- Built a VGG-style CNN and training framework from scratch in C++/CUDA/Eigen with no ML dependencies, driving six kernel generations under CUDA-event timing with bit-identical output across all of them.

VistaSR | Python, PyTorch, Diffusion Models

github.com/fordatab/sr

- Developed a diffusion-based super-resolution model (SR3) enabling 2x upscaling of landscape images, outperforming Real-ESRGAN in qualitative evaluations.
- Engineered a refined U-Net architecture with residual blocks and self-attention, enabling efficient training on mid-tier hardware to iteratively denoise inputs.
- Conducted quantitative assessments using PSNR/SSIM metrics and visual comparisons against FSRCNN, resulting in a robust solution for real-world applications.

TriStage Compiler | Java, ANTLR

github.com/fordatab/tristage

- Built a compiler for a custom imperative language; implemented an ANTLR front-end for precise tokenization, error reporting, and LL(1) grammar resolution.
- Optimized the middle-end via static analysis and dead code elimination, reducing dynamic instruction counts by 50% over unoptimized IR.
- Engineered a MIPS32 back-end with intra-block greedy register allocation and liveness analysis, yielding 30-60% fewer memory loads.

Skills

- **Languages:** C++, CUDA, Python, Java, JavaScript, TypeScript, SQL, Solidity
- **Front-End:** React, React Native, Redux, Tailwind CSS
- **Back-End:** Node.js, Express.js, Supabase, PostgreSQL, Vector Embeddings
- **Cloud & Tools:** GCP (Vertex AI, Cloud Run, BigQuery), Docker, GitHub Actions, Git, AWS S3